

SV

From Euler characteristic to Betti ~~number~~ sum

We need to estimate $B(X; \mathbb{Q}_l) = \sum h^i(X; \mathbb{Q}_l)$, we give a upper bound from the upper bound of $\chi(X; \mathbb{Q}_l) = \sum (-1)^i h^i(X; \mathbb{Q}_l)$, Katz prove it by weak Lefschetz thm.

Then as χ can be computed by index thm and characteristic classes, we can estimate B ; All cohomology here are étale.

Q. Why we need B than χ enough? What can B do but χ can't? We answer it at the end of this note.

Thm. Let $V \subset \mathbb{A}^N$ closed scheme, i.e. affine, $|\chi_c(V; \mathbb{Q}_l)| \leq E_{N,r,d}$, V defined by r polynomial's zero, all $\deg \leq d$, then:

① If V is smooth and connected (or $\dim V = 0$), then $B_c(V; \mathbb{Q}_l)$

$$\leq A_{N,r,d} := E_{N,r,d} + 2 + 2 \sum_{1 \leq h \leq N-1} E_{h,r,d}$$

② Without V smooth and connected, then $B_c(V; \mathbb{Q}_l) \leq B_{N,r,d}$

$$:= 1 + \sum_{\emptyset \neq S \subset \{1, \dots, r\}} A(N+1, 1, |S|d \cdot (\#S))$$

Rk. For V smooth, we can apply Poincaré dual of each component get $B_c(V; \mathbb{Q}_l) = B(V; \mathbb{Q}_l)$; For V projective, also.

And in projective case, set $X \subset \mathbb{P}^N$, $X = (X \cap \mathbb{A}^N) \sqcup (X \cap \mathbb{P}^{N-1})$

induction on N , apply ②, we have $\cong: V^N \sqcup X^{N-1}$

$$B_c(V; \mathbb{Q}_l) \leq 1 + \sum_{1 \leq n \leq N} B(n, r, d)$$

Pf of Thm. For ①, we use hyperplane section and weak Lefschetz, it's nontrivial; For ②, it's trivial, by choosing a "good cover" and use ~~long exact~~ spectral sequence.

① $\Rightarrow$ ②

Excision sequence;

$$\cdots \rightarrow H_c^{i-1}(\mathbb{A}^N; \mathbb{Q}_l) \rightarrow H_c^{i-1}(V; \mathbb{Q}_l) \rightarrow H_c^i(\mathbb{A}^N - V; \mathbb{Q}_l) \rightarrow \cdots$$

$\Rightarrow B_c(V; \mathbb{Q}_l) \leq B_c(\mathbb{A}^N; \mathbb{Q}_l) + B_c(\mathbb{A}^N - V; \mathbb{Q}_l)$ thus it left to prove

$$B_c(\mathbb{A}^N - V; \mathbb{Q}_l) \leq \sum A_{N+1,1,1+d \cdot \#S}$$

affine have the standard cover as the good cover



V defined by $F_1, \dots, F_r$, then $A^N - V = \bigcap_{i=1}^r (A^N - Z(F_i))$
 $= U(A^N - Z(F_i)) = U D(F_i)$

$U_i := D(F_i)$ form a good cover as $U_{ij} = D(F_i) \cap D(F_j) = D(F_i F_j)$

thus we have $E_1^{p,q} = \bigoplus_{\substack{\emptyset \neq S \subset \{1, \dots, r\} \\ S = \{i_1, \dots, i_p\}}} H^q(U_{i_1} \cap \dots \cap U_{i_p}; \mathbb{Q}_\ell) \Rightarrow E_\infty^{p,q} = H^{p+q}(A^N - V; \mathbb{Q}_\ell)$

$$\Rightarrow B_0(VA^N - V; \mathbb{Q}_W) \leq \sum_p \sum_s B(U_p \cap \dots \cap U_{ip}; \mathbb{Q}_W)$$

$$= \sum_{p=1}^n \sum_{s=1}^n B(D(F_p) - F_{ip}; \mathbb{Q}_W) =: \sum_{p,s} BCD(F_s; \mathbb{Q}_W)$$

$D(f_3) \subset \mathbb{A}^n - V \subset \mathbb{A}^n \subset \mathbb{A}^{n+1}$ is a locally closed hypersurface.

$D(f_3) = \sqrt{1 - z f_3}$ as the zero of $1 - z f_3 = 0$ is $f_3 \neq 0, z = 0$

thus $D(F_3) \subset \mathbb{A}^{N+1}$ is a smooth connected hypersurface.

$$\textcircled{1} \Rightarrow BCAN-V; (Q_U) \in \sum_{P.S} A_{N(t,1), 1 + \deg T_S} \leq \sum_{P.S} A_{N(t,1), 1 + p_d} \leq \sum_S A_{N(t,1), H_K}$$

PF of $\textcircled{1}$

Run induction on $n = \dim V$:

When $\dim V = 0$, $B_0(V; \mathbb{Q}) = h^0(V; \mathbb{Q}) = \chi(V; \mathbb{Q}) \leq E \leq A$;

~~Because~~ (Because latter we have $n-2$, we induct for $n \geq 2$, here induct)

When $\dim V = 1$, $B(V; \mathbb{Q}_U) = 1 + h^1(V; \mathbb{Q}_U)$

$$= 2 - \chi(V_j @ \mathbb{Q}) \leq E + 2 \leq A_j$$

When $n \geq 2$ and it holds for $(n-1)$ case, then we apply:

(Affine Weak Lefschetz) For general $H \subset \mathbb{A}^N$ hyperplane, $\dim H = n$

- $V(M)$ is still smooth, connected, $\dim = n-1$

- $H^i(V; \mathbb{Q}) \xrightarrow{j^*} H^i(V \cap H; \mathbb{Q})$ is injective for $i = n-1$

$(i: V \cap H \hookrightarrow V)$ isomorphic for $i \leq n-2$

~~Handwritten scribbles and symbols~~

$$\begin{aligned} \Rightarrow h^n(V; \mathbb{Q}_\ell) &= (-1)^n \chi(V; \mathbb{Q}_\ell) + (-1)^{n-1} \chi(V; \mathbb{Q}_\ell) + h^n(V; \mathbb{Q}_\ell) \\ &= (-1)^n \chi(V; \mathbb{Q}_\ell) + (-1)^{n-1} (\chi(V; \mathbb{Q}_\ell) - (-1)^n h^n(V; \mathbb{Q}_\ell)) \\ &= (-1)^n \chi(V; \mathbb{Q}_\ell) + (-1)^{n-1} (\sum_{0 \leq i \leq n-1} (-1)^i h^i(V; \mathbb{Q}_\ell)) \end{aligned}$$

$$h^i(V; \mathbb{Q}_\ell) = \begin{cases} h^i(V \cap H; \mathbb{Q}_\ell) & 0 \leq i \leq n-2 \\ \leq h^{n-1}(V \cap H; \mathbb{Q}_\ell); i = n-1 \end{cases}$$

$$\Rightarrow h^n(V; \mathbb{Q}) \leq (-1)^n \chi(V; \mathbb{Q}) + (-1)^{n-1} \chi(V \cap H; \mathbb{Q})$$

$$\Rightarrow B(V; \mathbb{Q}_\ell) = h^0(V; \mathbb{Q}_\ell) + \sum_{i=1}^{\dim V} h^i(V; \mathbb{Q}_\ell)$$

$$\begin{aligned} &= h^n(V; \mathbb{Q}_L) + \sum_{0 \leq i \leq n-1} B(V \cap H_i; \mathbb{Q}_L) \\ &= (-1)^n \chi(V; \mathbb{Q}_L) + (-1)^{n-1} \chi(V \cap H; \mathbb{Q}_L) + B(V \cap H; \mathbb{Q}_L) \\ &\stackrel{\text{induction}}{\leq} E_{n,r,d} + E_{n-1,r,d} + A_{n-1,r,d} = A_{n,r,d} \quad \square \end{aligned}$$

Application of B then X:

Recall the trivial bound of $|X(\mathbb{F}_q)| = \sum (-1)^i \text{tr}(\text{Frob} | H_i^c)$ (by trace

formulae we use: H^i_c (than stalk of sheaf) is given by Weil conj:

Prob has eigenvalue $\leq g^{\frac{1}{2}}$ on $H^{\frac{1}{2}}$ & size of $H^{\frac{1}{2}}$ is $h^{\frac{1}{2}}$

$$\Rightarrow |X(\mathbb{F}_q)| \leq \sum h_i^c(X) \cdot q^{\frac{v_i}{2}}, \text{ set } d = \dim X$$

$$\Rightarrow |X(F_q) - q^{\frac{d}{2}}| \leq \sum_{i=1}^d h_i(X) \cdot q^{\frac{i}{2}} \leq B_C(X; \mathbb{Q}) \cdot q^{d-\frac{1}{2}}$$

For X is curve $\Rightarrow |X(\frac{1}{\sqrt{2}}) - (8+1)| \leq 2g\sqrt{g}$ is the quadratic cancellation

$$\langle B_c(X; \mathbb{Q}_\ell) \sqrt{q} = (2g+2) \sqrt{q}$$

but $x = 2 - 2g$ can do nothing (or even $|x| = 2g - 2$)

Similarly, it works for other arithmetic problem, e.g. the exponential

sum, by replace $\mathbb{Q}_L$ by $\mathbb{Z}_p$ relatively, ~~$\mathbb{Q}_L$~~ and $B_c(X; \mathbb{Z}_p)$ is used. But $\mathbb{Z}_p \cong \bigoplus \mathbb{Q}_L$, thus the estimate of $\chi_c(X; \mathbb{Z}_p)$ is similar as $\chi_c(X; \mathbb{Q}_L)$.